Peddi Manognya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.25
Sri Chaitanya Junior College	Class XII (TSBIE)	2020	97.60%
Oxford High School	Class X (SSC)	2018	10.00%

PUBLICATIONS

Peddi Manognya, Lokendra Mandloi, Joshi Sayali Shripad, Sandipan Dandapat. "*Values Are Not Enough: Situational Modifiers Shape Moral Decisions in LLMs and Humans*." Conference on Empirical Methods in Natural Language Processing (EMNLP 2026), Budapest, Hungary. **Accepted to Findings**.

PROJECTS

VISTA: Value Informed Scenario based Task Agent [Github]

- Studied the effect of changes in decisions with respect to a person's value system and scenarios.
- Extended the project to study the effect of scenario modifiers.
- Evaluated 7 LLMs (Gemma 4, Qwen 2.5, and LLaMA 3.1, Claude Haiku 4.5, Claude Sonnet 5, GPT-4.1-mini, and GPT-5-mini) using a novel decision-flip metric and human comparison study.

Client-Aware Radio Resource Management (RRM) in Wi-Fi [Github]

- Developed a client-aware RRM strategy that adapts Wi-Fi resource allocation based on client channel conditions and traffic demand to improve overall network performance.
- Designed adaptive association and scheduling logic to improve airtime fairness and overall network throughput.
- Simulated and evaluated performance under varying client densities and interference scenarios.
- Benchmarked solution against baseline allocation schemes, demonstrating improved QoS metrics.

TransactiWar - Secure Web Application & Offensive Security Assessment [Github]

- Developed a full-stack banking web application with account creation, deposit, and withdrawal features using PHP.
- Implemented defenses against major web vulnerabilities including SQL Injection, XSS, CSRF, LFI, and RFI.
- Designed secure session management and input validation to ensure safe user transactions.
- Delivered a robust web security prototype showcasing end-to-end protection for financial applications.

SDR-Based End-to-End 4G/5G OTA Setup

- Deployed and configured an end-to-end 4G/5G cellular setup using USRP B210 and OpenAirInterface in a Linux environment, enabling over-the-air communication between base station and UE.
- Established over-the-air communication between UE and base station, validating PHY and MAC layer functionality.
- Analyzed synchronization, RF configuration, and throughput behavior using OAI logs and system diagnostics.
- Conducted controlled experiments to study link stability and performance under different radio configurations.

SKILLS

Languages: C, C++, Python, Java, SQL

Machine Learning & AI: Scikit-learn, PyTorch, Deep Learning, Machine Learning

Networking & Systems: Wireshark, TCPdump, ns-3, OpenAirInterface, srsRAN

Tools & Platforms: Linux, Docker, Git/GitHub, Jupyter Notebook, Google Colab

Wireless & Networking: LTE/5G basics, Handover mechanisms (A3, A2-A4), O-RAN (xApps, RIC concepts), Wi-Fi

RRM, ns-3 simulations, basic network traffic analysis, OpenAirInterface, srsRAN

RELEVANT COURSES

Advanced Computer Networks | Foundations of Machine Learning | Advaned Data Structures and Algorithms | Topics in Deep Learning | Networked Wireless Systems | Cryptography | Network Security | Fraud Analytics using Predictive and Social Network Techniques | Wireless Networks and Security

POSITIONS OF RESPONSIBILITY

- **Placement Coordinator:** managed student-recruiter communication and helped organize placement activities.
- **Volunteer, ACM ARCS 2026:** supported event coordination and execution of technical sessions.

EXTRACURRICULAR

- Organized workshops and technical sessions as part of CreScencE-2k23 (department fest).
- Volunteered with the NSS Cell, JNTUH-UCEJ, organizing Swachh Bharat Abhiyaan, health camps, and technology awareness drives in villages as part of an NSS special drive.

SCHOLASTIC ACHIEVEMENTS

- Selected as Trainee Scientific Officer (TSO) at BARC through OCES/DGFS-2024.
- 1st at district level - Mathematics Olympiad (2017).